Microwave Engineering- ECA1308

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Experiment 2A: Transmission-Line/TN-Port S-Parameters: Model a 50 2 microwave two-port network of length 50 mm and operate it from 8 to 12 GHz with a 10 GHz center frequency. Set both ports to 50 and obtain $11, S12, S21 and S22. At 10 GHz calculate return loss RL-20log10/S11, insertion loss IL--20log10 S21 and transmission efficiency |S21F×100%. Compare the simulated values with the ideal 500 matched reference.

Aim: To model a 50 Ω microwave two-port transmission line of length 50 mm in MATLAB, obtain the S-parameters S_{11}, S_{12}, S_{21}, S_{22}\) over 8–12 GHz, and calculate the return loss, insertion loss, and transmission efficiency at 10 GHz.

Software Required:MATLAB

Procedure — MATLAB Process

1. Open **MATLAB** and create a new script using **Home → New Script**.
2. Define the input parameters: center frequency \(f_0=10\) GHz, reference impedance \(Z_0=50\ \Omega\), and transmission-line length \(L=50\) mm.
3. Define the frequency sweep from **8 GHz to 12 GHz** using the MATLAB linspace() function.
4. Calculate the phase constant \(\beta=2\pi f/c\), assuming an ideal lossless transmission line with propagation velocity equal to the speed of light.
5. Calculate the transmission-line **ABCD parameters** \(A,B,C,D\) using the line length and characteristic impedance.
6. Convert the ABCD parameters into the four S-parameters \(S_{11},S_{12},S_{21},S_{22}\).
7. Find the S-parameter values specifically at **10 GHz** from the frequency sweep.
8. Calculate the **return loss**, **insertion loss**, and **transmission efficiency** using \(RL=-20\log_{10}|S_{11}|\),

\(IL=-20\log_{10}|S_{21}|\), and
\(\eta=|S_{21}|^2\times100\%\).

9. Run the MATLAB program, observe the numerical results, and plot the S-parameters versus frequency to analyze the transmission-line performance.

OUTPUT:

FIG:1

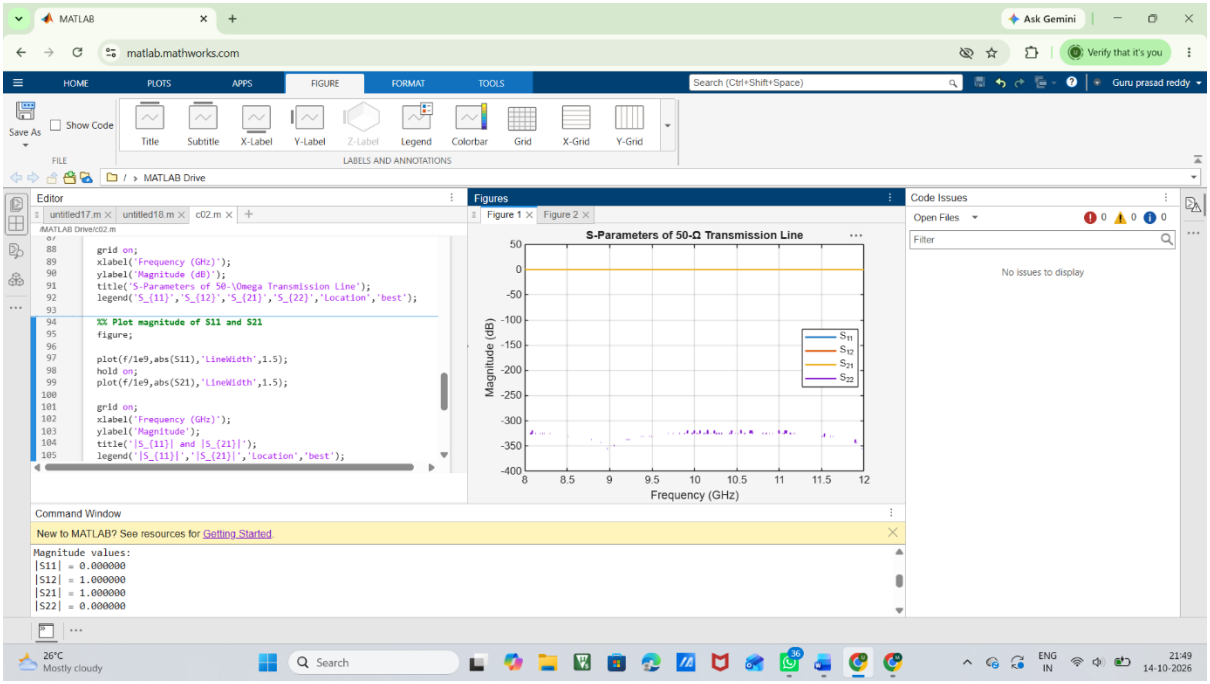
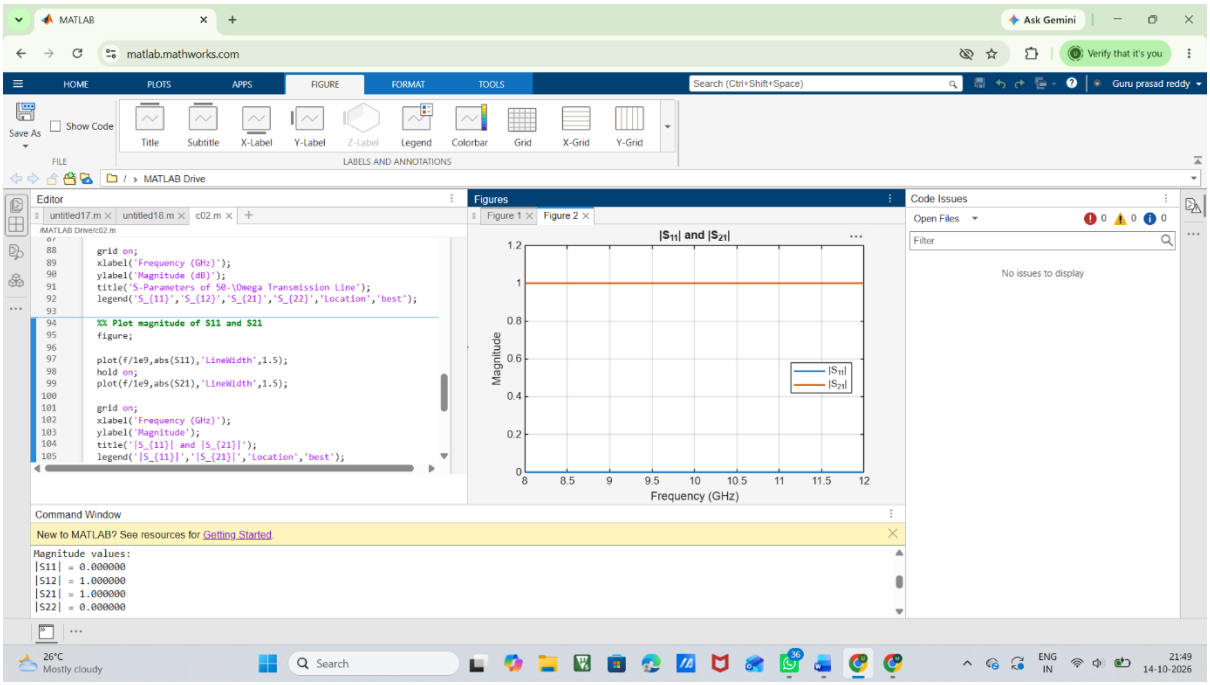


FIG:B



Result —

1. The S-parameters \(S_{11}, S_{12}, S_{21}\), and \(S_{22}\) were successfully obtained for the 8–12 GHz frequency range.
2. At 10 GHz, the ideal 50 Ω line gives \(|S_{11}|\approx0\) and \(|S_{21}|\approx1\).
3. The return loss is theoretically very high (infinite for an ideal match), indicating excellent matching.
4. The insertion loss is theoretically **0 dB**, indicating lossless transmission.
5. The transmission efficiency is approximately **100%**, confirming ideal power transfer.